



HK IMO Training 2021-2022
Problem Set 17
For 12 February 2022



Problem 65. Let p and q be prime numbers such that $p + q^2$ is a perfect square. Prove that $p^2 + q^n$ is not a perfect square for any positive integer n .

Problem 66. Let n be a positive integer. Let A be the number of tuples $(a_1, a_2, \dots, a_r)$ of positive integers such that $a_1 + a_2 + \dots + a_r = n$, $a_{r-1} > a_r$ and $a_i > a_{i+1} + a_{i+2}$ for $i = 1, 2, \dots, r-2$ (where r can be any positive integer). Define the sequence $\{x_n\}$ by $x_1 = 1$, $x_2 = 2$ and $x_n = x_{n-1} + x_{n-2} + 1$ for $n \geq 3$. Let B be the number of tuples $(b_1, b_2, \dots, b_s)$ of positive integers such that $b_1 + b_2 + \dots + b_s = n$, $b_1 \geq b_2 \geq \dots \geq b_s$, and there exists k for which the set $\{b_1, b_2, \dots, b_s\}$ is the same as the set $\{x_1, x_2, \dots, x_k\}$ (where s can be any positive integer). Prove that $A = B$.

Problem 67. Let $a_1, a_2, \dots, a_n$ be given positive real numbers. Prove that there exist positive real numbers $b_1, b_2, \dots, b_n$ such that $b_i \geq a_i$ for each i , $\frac{b_i}{b_j}$ is an integer whenever $b_i \geq b_j$, and $b_1 b_2 \dots b_n \leq 2^{\frac{n-1}{2}} a_1 a_2 \dots a_n$.

Problem 68. Let $ABCD$ be a convex quadrilateral whose sides are not parallel and whose vertices are not concyclic. The lines AB and CD meet at E . The circumcircles of $\triangle ADE$ and $\triangle BCE$ meet again at P . The four internal angle bisectors of $ABCD$ enclose a cyclic quadrilateral with circumcentre I . The four external angle bisectors of $ABCD$ enclose a cyclic quadrilateral with circumcentre J . Prove that P, I, J are collinear.